NEET TEST HUB

NTH-PREMIUM TEST SERIES 02

Date – 14 September 2025	Duration – 3 Hours	Marks – 720

Physics :- Motion in a Straight Line, Motion in a Plane

Chemistry :- Thermodynamics

Biology :- Cell Cycle & Cell Division, The Living World,
Structural Organisation in Animals, Breathing & Exchange of Gases

Instruction

1. The test is of 3 hours duration.

2. The test booklet consists of 200 questions. The maximum mark is 720.

3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,

4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).

5. There is only one correct response for each question.

6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.

7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.

8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

Physics

1. The friction of the air causes vertical retardation equal to 10% of the acceleration due to gravity. The maximum height will be decreased by:

(Take $g = 10$ ms^{-2})

1.	8%	2.	9%
3.	10%	4.	11%

2. A car starts at point X. It travels 3.0 km due east, then 4.0 km due south, then 6.0 km due west, and finally 8.0 km due north. How far away is the car from point X when it has reached the end of this journey? (Assume that all distances moved are on a flat horizontal surface and that point X is on the equator. You may ignore any curvature of the Earth).
1. 5.0 km
2. 21.0 km
3. 10.0 km
4. 7.0 km

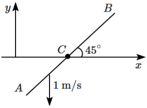
3. The net force acting on a projectile at the highest point of its trajectory, moving with a speed v, at a height H is:
(m is the mass of the projectile)
1. zero
2. mg
3. $mg + \dfrac{mv^2}{H}$
4. $mg - \dfrac{mv^2}{H}$

4. A particle moves around a circle with a unique uniform speed in each revolution. After the first revolution and during the 2nd revolution: its speed doubles; and during the 3rd revolution, its speed becomes 3 times the initial speed and so on. The time for the 1st revolution is 12 sec. The average time per revolution, for the first four revolutions, is:
1. 4.8 s
2. 9.6 s
3. 6.25 s
4. 6 s

5. A ball, thrown vertically upward, is observed to move upward with a speed v_1 at time t_1 and a speed v_2, downward, at time t_2.
The acceleration of the ball is (downward):

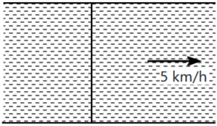
1.	$\dfrac{v_2 - v_1}{t_2 - t_1}$	2.	$\dfrac{v_2 + v_1}{t_2 - t_1}$
3.	$\dfrac{v_2 - v_1}{t_2 + t_1}$	4.	$\dfrac{v_2 + v_1}{t_2 + t_1}$

6. The line AB makes a 45° angle with the x-axis, but it moves along the negative y-axis with a speed of 1 m/s. The velocity, of the intersection (C) of AB with x-axis, is:



1.	1 m/s along the positive x-axis
2.	1 m/s along the negative x-axis
3.	$\dfrac{1}{\sqrt{2}}$ m/s along the positive x-axis
4.	$\dfrac{1}{\sqrt{2}}$ m/s along the negative x-axis

7. In the figure shown, a river of width 4 km is flowing with a speed of 5 km/h. A swimmer whose swimming speed relative to the water is 4 km/h, starts swimming from point A on the bank. On the other bank, B is a point that is directly opposite to A. At what angle with the downstream the man should swim so that he reaches point B directly?



1. 90°
2. 120°
3. 150°
4. Not possible

8. Two bullets are fired horizontally and simultaneously towards each other from the rooftops of two buildings (building being $100 \mathrm{~m}$ apart and being of the same height of $200 \mathrm{~m}$) with the same velocity of $25 \mathrm{~m/s}$. When and where will the two bullets collide? $\left(g=10 \mathrm{~m/s}^2\right)$

1.	After 2 s at a height of 180 m
2.	After 2 s at a height of 20 m
3.	After 4 s at a height of 120 m
4.	They will not collide.

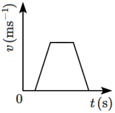
9. Two particles are projected in the air at the same speed u at an angle θ_1 and θ_2 (θ_1 and θ_2 are acute angles) to the horizontal, respectively. If the maximum height reached by the first particle is greater than that of the second, then which of the following is correct? (T_1 and T_2 are the times of flight of the two particles respectively)
1. $\theta_1>\theta_2$
2. $\theta_1=\theta_2$
3. $T_1<T_2$
4. $T_1=T_2$

10. Two trains each of length $100 \mathrm{~m}$, are running on parallel tracks. One train overtakes the other in 20 s and they cross each other in 10 s. What are the speeds of the two trains?
1. $10 \mathrm{~m/s}$, $20 \mathrm{~m/s}$
2. $20 \mathrm{~m/s}$, $30 \mathrm{~m/s}$,
3. $15 \mathrm{~m/s}$, $5 \mathrm{~m/s}$,
4. $25 \mathrm{~m/s}$, $15 \mathrm{~m/s}$,

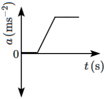
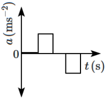
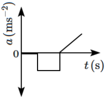
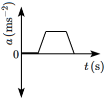
11. A particle moving in a uniform circular motion of radius 1 m has velocity $3\hat{j} \mathrm{~m/s}$ at point B. What are the velocity $(\vec{v})$and acceleration $(\vec{a})$ at diametrically opposite point A?

1.	$\vec{v}_A=3\hat{j} \mathrm{~m/s} ; \vec{a}_A=-9\hat{i} \mathrm{~m/s}^2$
2.	$\vec{v}_A=-3\hat{j} \mathrm{~m/s} ; \vec{a}_A=9\hat{i} \mathrm{~m/s}^2$
3.	$\vec{v}_A=-3\hat{i} \mathrm{~m/s} ; \vec{a}_A=9\hat{j} \mathrm{~m/s}^2$
4.	$\vec{v}_A=3\hat{i} \mathrm{~m/s} ; \vec{a}_A=9\hat{j} \mathrm{~m/s}^2$

12. The velocity (v)-time (t) plot of the motion of a body is shown below:

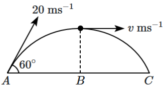


The acceleration (a)-time (t) graph that best suits this motion is:

1.		2.	
3.		4.	

13. A car is going up a slight slope decelerating at 0.1 m/s^2. It comes to a stop after going for 5 s. What was its initial velocity?
1. $0.02 \mathrm{~m/s}$
2. $0.25 \mathrm{~m/s}$
3. $0.5 \mathrm{~m/s}$
4. $1.0 \mathrm{~m/s}$

14. A ball is projected from point A with velocity 20 ms^{-1} at an angle 60° to the horizontal direction. At the highest point B of the path (as shown in figure), the velocity v (in ms^{-1}) of the ball will be:

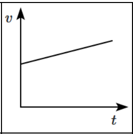
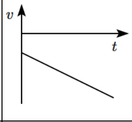
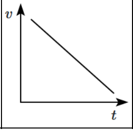
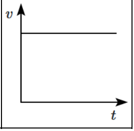


1.	20	2.	$10\sqrt{3}$
3.	zero	4.	10

15. The position vector of a particle is expressed as: $\overrightarrow{r}(t) = (8t\hat{i} + 5t^2\hat{j} + 6\hat{k})$. Which of the following statements correctly describes the direction of the particle's acceleration?

1.	It is directed along the positive y-axis.
2.	It is directed along the positive x-axis.
3.	It is equally inclined to the x and y-axis.
4.	It is directed along the positive z-axis.

16. A car moving in a positive x-direction with uniform positive acceleration. Which one of the following velocity-time graphs correctly represents the motion of the car?

1.		2.	
3.		4.	

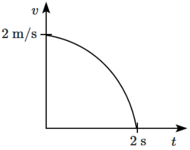
17. A particle is moving along a straight line such that its position depends on time as $x = 1 - at + bt^2$, where $a = 2\ \mathrm{m/s}$, $b = 1\ \mathrm{m/s^2}$. The distance covered by the particle during the first 3 seconds from start of the motion will be:

1.	2 m	2.	5 m
3.	7 m	4.	4 m

18. The velocity of projection of an oblique projectile is $\vec{v} = \left(3\hat{i} + 2\hat{j}\right)$ ms^{-1}. The speed of the projectile at the highest point of the trajectory is:

1. 3 ms^{-1}
2. 2 ms^{-1}
3. 1 ms^{-1}
4. zero

19. The velocity-time graph of a particle, moving along a straight time, is shown in the figure. The curve, when plotted, takes the form of a 'circle'. The magnitude of the average acceleration of the particle is:



1.	1 m/s^2
2.	2 m/s^2
3.	less than 1 m/s^2
4.	greater than 2 m/s^2

20. A particle moving with a speed v changes direction by an angle θ, without a change in the speed.

(A)	The change in the magnitude of its velocity is zero.
(B)	The change in the magnitude of its velocity is $2v\sin(\theta/2)$.
(C)	The magnitude of the change in its velocity is $2v\sin(\theta/2)$.
(D)	The magnitude of the change in its velocity is $v(1-\cos\theta)$.

Choose the correct option from the options given below:

1.	**(A), (B)** and **(C)** only
2.	**(A)** and **(C)** only
3.	**(B)** and **(D)** only
4.	**(B), (C)** and **(D)** only

21. A clock has a continuously moving second's hand of 0.1 m length. The average acceleration of the tip of the second hand (in units of ms^{-2}) is of the order of:

1. 10^{-3}
2. 10^{-4}
3. 10^{-1}
4. 10^{-2}

22. Two cars A & B move from town X to town Y: both accelerating uniformly from rest to a maximum speed of $40\ \mathrm{km/h}$ and immediately, decelerating uniformly to a stop at Y. The distance between X & Y is $120\ \mathrm{km}$. For car A,the acceleration & deceleration have the same magnitude; for car B, the acceleration is twice the deceleration. The times taken by A & B are t_A & t_B. Then:

1.	$t_A = 2t_B$	2.	$t_B = 2t_A$
3.	$3t_A = 2t_B$	4.	$t_A = t_B$

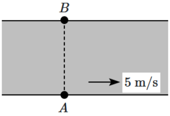
23. A particle is moving such that its position coordinates (x, y) are (2 m, 3 m) at time $t = 0$, (6 m,7 m) at time $t = 2$ s, and (13 m, 14 m) at time $t = 5$ s. The average velocity vector $\vec{v}_{avg}$ from $t = 0$ to $t = 5$ s is:

1. $\frac{1}{5}(13\hat{i} + 14\hat{j})$
2. $\frac{7}{3}(\hat{i} + \hat{j})$
3. $2(\hat{i} + \hat{j})$
4. $\frac{11}{5}(\hat{i} + \hat{j})$

24. Two cars P and Q start from a point at the same time in a straight line and their positions are represented by; $x_p(t) = at + bt^2$ and $x_Q(t) = ft - t^2$. At what time do the cars have the same velocity?

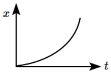
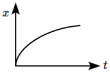
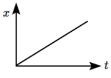
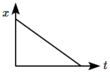
1.	$\frac{a-f}{1+b}$	2.	$\frac{a+f}{2(b-1)}$
3.	$\frac{a+f}{2(b+1)}$	4.	$\frac{f-a}{2(1+b)}$

25. A man swims directly across a river, where his destination (B) is directly across the river from his starting point (A). The speed of flow of the river water is $5\ \mathrm{m/s}$. The speed of (effort of) the man w.r.t., the river water:



1. is $5\ \mathrm{m/s}$
2. is less than $5\ \mathrm{m/s}$
3. is greater than $5\ \mathrm{m/s}$
4. can be any of the above

26. The position-time $(x\text{-}t)$ graph for positive acceleration is:

1.		2.	
3.		4.	

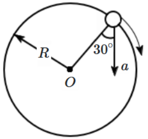
27. A mass is projected from the ground with a certain velocity, at an angle with the horizontal. Neglecting air resistance, we can say that its speed will be:

1.	constant during the flight
2.	decreasing continuously during the flight
3.	minimum at the highest point
4.	maximum at the highest point

28. $\hat{i}$ and $\hat{j}$ are unit vectors along the x, and y-axis respectively. What is the magnitude and direction of the vector $(\hat{i} - \hat{j})$?

1.	$\sqrt{2}$, 45° with the x-axis.
2.	$\sqrt{2}$, -45° with the x-axis.
3.	$\frac{1}{\sqrt{2}}$, 60° with the x-axis.
4.	$\frac{1}{\sqrt{2}}$, -60° with the x-axis.

29. In the given figure, $a = 15$ m/s^2 represents the total acceleration of a particle moving in the clockwise direction in a circle of radius $R = 2.5$ m at a given instant of time. The speed of the particle is:



1. 4.5 m/s
2. 5.0 m/s
3. 5.7 m/s
4. 6.2 m/s

30. Which of the following motions of an object always results in zero acceleration?

1.	Any motion in a straight line.
2.	Projectile motion.
3.	Any motion in a circle.
4.	None of the motions above guarantees zero acceleration.

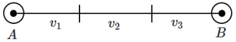
31. An elephant pulls a large tree trunk, 4 km in a direction 50° east of north. What are the components of the elephant's displacement in the easterly and northerly directions?
(Given, $\sin 50^\circ = 0.766$ and $\cos 50^\circ = 0.642$)

1.	3.06 km east and 2.57 km north
2.	2.57 km east and 3.06 km north
3.	3.94 km east and 0.69 km north
4.	0.69 km east and 3.94 km north

32. The equations of motion for a projectile are given as $x = 36t$ and $2y = 96t - 9.8t^2$, where x and y are in metres and t is in seconds. What is the angle of projection (measured from the x-axis)?

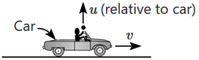
1.	$\sin^{-1}\left(\frac{4}{5}\right)$	2.	$\sin^{-1}\left(\frac{3}{5}\right)$
3.	$\sin^{-1}\left(\frac{4}{3}\right)$	4.	$\sin^{-1}\left(\frac{3}{4}\right)$

33. A car travels a distance AB in three equal segments: the first third at a velocity of v_1 m/s, the second third at v_2 m/s, and the final third at v_3 m/s. Given that $v_3 = 3v_1$, $v_2 = 2v_1$ and $v_1 = 11$ m/s, the average velocity of the car is:



1.	12 m/s	2.	18 m/s
3.	16 m/s	4.	11 m/s

34. A ball is thrown vertically upward with speed u, by a man in a car, which is moving forward horizontally with a speed v. The angle of throw (i.e., with the horizontal) as seen from the ground is θ. Then:



1. $v \cos\theta = u \sin\theta$
2. $v \sin\theta = u \cos\theta$
3. $v \tan\theta = u \cot\theta$
4. $v \cot\theta = u \tan\theta$

35. The numerical ratio of average velocity to average speed is:

1.	equal to one	2.	greater than one
3.	less than one	4.	equal to or less than one

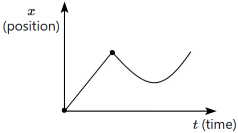
36. A man runs along a horizontal road holding his umbrella vertical in order to afford maximum protection from the rain. The rain is actually:

1.	falling vertical
2.	coming from the front of the man
3.	coming from the back of the man
4.	either of 1, 2, or 3

37. The velocity of a particle at any instant is given by the equation $\vec{v} = (2t\hat{i} + 3t^2\hat{j})$ m/s, and the radius of the curvature of the path is 2 m. The centripetal acceleration of the particle at $t = 2$ s will be:

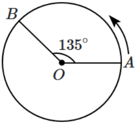
1. 80 m/s^2
2. 160 m/s^2
3. 40 m/s^2
4. 100 m/s^2

38. Which of the following statements is true about the motion depicted in the diagram?



1.	The acceleration is constant and non-zero.
2.	The velocity changes suddenly during the motion.
3.	The velocity is positive throughout.
4.	All of the above are true.

39. A person moves from point A to point B along a circular arc that subtends an angle of 135° at the centre O, as shown in the figure. The length of the arc AB is 60 m. If $\cos 135^\circ = -0.7$, the magnitude of the displacement of the person is:



1. 42 m
2. 47 m
3. 19 m
4. 40 m

40. A ball of mass 5 grams is thrown downward from point A with a certain initial velocity v_0 towards a hard floor. Without any loss of energy, it rebounds from the floor to a point 10 m above the original level. Thus, the initial velocity v_0 of the ball is:
1. 10 m/s
2. 50 m/s
3. 14 m/s
4. 28 m/s

41. A body is dropped from a height of 100 m. At what height the velocity of the body will be equal to one-half of the velocity when it hits the ground?

1.	25 m	2.	55 m
3.	65 m	4.	75 m

42. A body sliding on a smooth inclined plane requires 4 s to reach the bottom starting from rest at the top. How much time does it take to cover one-fourth the distance starting from rest at the top?
1. 1 s
2. 2 s
3. 4 s
4. 16 s

43. A ball is thrown at an angle θ_0 above the horizontal, and follows the parabolic path taken by a projectile. Let its speed be v when its trajectory makes an angle θ with the horizontal. Assuming A to be a constant,
1. $v = A\cos\theta$
2. $v = A\sin\theta$
3. $v = A\tan\theta$
4. $v = A\sec\theta$

44. A ball is thrown vertically upward with an initial speed of 20 m/s. It reaches a height of 15 m above the ground, after a time of: (take $g = 10\text{ m/s}^2$)
1. 1 s
2. 3 s
3. 1 s or 3 s
4. 2 s

45. If the range of a gun which fires a shell with muzzle speed v, is R, then the angle of elevation of the gun is:

1.	$\cos^{-1}\left(\dfrac{v^2}{Rg}\right)$	2.	$\cos^{-1}\left(\dfrac{Rg}{v^2}\right)$
3.	$\dfrac{1}{2}\sin^{-1}\left(\dfrac{v^2}{Rg}\right)$	4.	$\dfrac{1}{2}\sin^{-1}\left(\dfrac{Rg}{v^2}\right)$

Chemistry

46. Choose the correct statement for the work done in the expansion and heat absorbed or released when 5 liters of an ideal gas at 10 atmospheric pressure isothermally expands into a vacuum until the volume is 15 liters:

1.	Both the heat and work done will be greater than zero.
2.	Heat absorbed will be less than zero and work done will be positive.
3.	Work done will be zero and heat absorbed or evolved will also be zero.
4.	Work done will be greater than zero and heat absorbed will remain zero.

47. For the reaction, $2Cl(g) \rightarrow Cl_2(g)$, the correct option is:
1. $\Delta_r H > 0$ and $\Delta_r S < 0$
2. $\Delta_r H < 0$ and $\Delta_r S > 0$
3. $\Delta_r H < 0$ and $\Delta_r S < 0$
4. $\Delta_r H > 0$ and $\Delta_r S > 0$

48. Lattice enthalpy and enthalpy of solution of NaCl are 788 kJ mol^{-1} and 4 kJ mol^{-1}, respectively. The hydration enthalpy of NaCl will be:
1. -780 kJ mol^{-1}
2. 780 kJ mol^{-1}
3. -784 kJ mol^{-1}
4. 784 kJ mol^{-1}

49. The enthalpy of vaporisation of CCl_4 is 30.5 kJ mol^{-1}. The heat required for the vaporisation of 284 g of CCl_4 at constant pressure is:
(Molar mass of $CCl_4 = 154$ g mol^{-1})
1. 56.25 kJ
2. 36.25 kJ
3. 26.55 kJ
4. 62.54 kJ

50. Consider the given two statements:

Assertion (A):	The temperature of a gas does not change when it undergoes an adiabatic expansion.
Reason (R):	During an adiabatic process, the container should be a perfect conductor.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

51. Calculate the standard enthalpy of reaction for the following reaction using the given enthalpies of reaction:
$3Co(s) + 2O_2(g) \rightarrow Co_3O_4(s)$
Given:
$2Co(s) + O_2(g) \rightarrow 2CoO(s); \Delta H_1^\circ = -475.8 \text{ kJ}$
$6CoO(s) + O_2(g) \rightarrow 2Co_3O_4(s); \Delta H_2^\circ = -355.0 \text{ kJ}$
1. -891.2 kJ
2. -120.8 kJ
3. $+891.2$ kJ
4. -830.8 kJ

52. Mark the correct statement regarding entropy.

1.	At 0°C, the entropy of a perfectly crystalline substance is taken to be zero.
2.	At absolute zero temperature, the entropy of a perfectly crystalline substance is positive.
3.	At absolute zero temperature, the entropy of all crystalline substances is taken to be zero.
4.	At absolute zero temperature, the entropy of a perfectly crystalline substance is taken to be zero.

53. A sample containing 1.0 mol of an ideal gas undergoes an isothermal, reversible expansion to ten times its original volume in two separate experiments. The expansion is carried out at temperatures of 300 K and 600 K, respectively.
The correct statements among the following are:

(a)	Work done at 600 K is 20 times the work done at 300 K.
(b)	Work done at 300 K is twice the work done at 600 K.
(c)	Work done at 600 K is twice the work done at 300 K.
(d)	$\Delta U = 0$ in both cases.

1. **(a)** and **(b)**
2. **(b)** and **(c)**
3. **(c)** and **(d)**
4. **(a)** and **(d)**

54. Given the following standard heats of formation:
$\Delta_f H^\ominus P_4O_{10}(s) = -3110 \text{ kJ/mol}$
$\Delta_f H^\ominus H_2O(l) = -286 \text{ kJ/mol}$
$\Delta_f H^\ominus H_3PO_4(s) = -1279 \text{ kJ/mol}$
Calculate the change in enthalpy $(\Delta_r H^\ominus)$ for the following reaction:
$P_4O_{10}(s) + 6H_2O(l) \rightarrow 4H_3PO_4(s)$

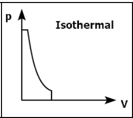
1.	-290 kJ	2.	-1128 kJ
3.	-2117 kJ	4.	-3547 kJ

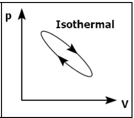
55. For the spontaneous process $H_2O_{(l)} \rightarrow H_2O_{(s)}$ at 268 K, the correct option is:
1. $\Delta S_{sys} < 0$
2. $\Delta S_{sys} > 0$
3. $\Delta S_{surr} < 0$
4. $\Delta S_{sys} = -\Delta S_{surr}$

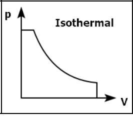
56. Choose the incorrect statement out of the following:

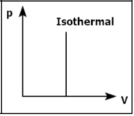
1.	$\Delta_r G = 0$ is the criteria for equilibrium.
2.	Second law of thermodynamics explains why spontaneous exothermic reactions are so common.
3.	ΔU can discriminate between reversible and irreversible processes.
4.	The heat released in the reaction $CaO(s) + CO_2(g) \rightarrow CaCO_3(s)$ is not the enthalpy of formation of $CaCO_3$

57. Which p-V curve among the following depicts the highest amount of work done?

1. 

2. 

3. 

4. 

58. Calculate the heat of the following reaction at constant pressure:
$F_2O(g) + H_2O(g) \rightarrow O_2(g) + 2HF(g)$
(Given: The heat of the formation of $F_2O(g), H_2O(g)$ and HF(g) is 5.5 kcal, -57 kcal, and -64 kcal respectively.)
1. -76.5 kcal
2. -82.5 kcal
3. -68.3 kcal
4. -95.6 kcal

59. Three thermochemical equations are given below:

(i)	$C_{(graphite)} + O_2(g) \rightarrow CO_2(g)$; $\Delta_r H° = x$ kJ mol^{-1}
(ii)	$C_{graphite} + 1/2\ O_2(g) \rightarrow CO(g)$; $\Delta_r H° = y$ kJ mol^{-1}
(iii)	$CO(g) + 1/2\ O_2(g) \rightarrow CO_2(g)$; $\Delta_r H° = z$ kJ mol^{-1}

Based on the above equations, find out which one of the relationships given below is correct :

1.	$z = x + y$	2.	$x = y + z$
3.	$y = 2z - x$	4.	$x = y - z$

60. The entropy change for the melting of 'x' moles of ice at 273 K and 1 atm pressure is 28.80 cal K^{-1}.
The value of 'x' is:
(Given: heat of fusion is 80 cal g^{-1})

1.	4.32 mol	2.	8.56 mol
3.	10.36 mol	4.	5.46 mol

61.

Assertion (A):	The heat absorbed during the isothermal expansion of an ideal gas against a vacuum is zero.
Reason (R):	The volume occupied by the molecules of an ideal gas is zero.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	**(A)** is False but **(R)** is True.

62. Which of the following is the standard enthalpy of formation of carbon monoxide (CO), if the standard enthalpies of combustion of carbon (C) and carbon monoxide (CO) respectively are −393.5 kJ mol^{-1} and −283 kJ mol^{-1}?
1. 110.5 kJ
2. 676.5 kJ
3. -676.5 kJ
4. -110.5 kJ

63. Mark the correct observation regarding the entropy of freezing of water in a glass.

1.	ΔS_{system} decreases but $\Delta S_{surroundings}$ remains the same
2.	ΔS_{system} increases but $\Delta S_{surroundings}$ decreases
3.	ΔS_{system} decreases but $\Delta S_{surroundings}$ increases
4.	ΔS_{system} decreases but $\Delta S_{surroundings}$ also decreases

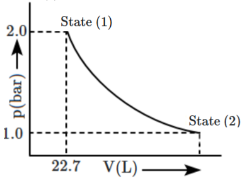
64. Given below are two statements:

Statement I:	A process or change is said to be reversible if it is brought out in such a way that the process can, at any point, be reversed by an infinitesimal change.
Statement II:	A reversible process proceeds infinitely slowly through a series of equilibrium states such that the system and its surroundings are always in near equilibrium with each other.

In light of the above statements, choose the correct answer from the options given below:

1.	Both **statement I** and **statement II** are correct.
2.	Both **statement I** and **statement II** are incorrect.
3.	**Statement I** is correct but **statement II** is incorrect.
4.	**Statement I** is incorrect and **statement II** is correct.

65. 1.0 mol of a monoatomic ideal gas is expanded from state (1) to state (2) as shown in the figure. What is the work done during the expansion of the gas from state (1) to state (2) at 298 K?



1. $W = -1717.46$ J
2. $W = -1618.52$ J
3. $W = +1717.46$ J
4. $W = -1200.5$ J

66. A system that can neither exchange matter nor energy with the surroundings is classified as:
1. Open system
2. Isolated system
3. Closed system
4. Both (1) & (2)

67. The correct relation for the work done in isothermal expansion will be:
1. $w_{reversible} < w_{irreversible}$
2. $w_{reversible} > w_{irreversible}$
3. $w_{reversible} = w_{irreversible}$
4. All of the above

68. Given the following reaction:
$\frac{1}{2}H_2 + \frac{1}{2}Cl_2 \rightarrow HCl$
$\Delta H_f(HCl) = -93\ kJ/mol$
$B.E\ (H_2) = 434\ kJ/mol,\ B.E\ (Cl_2) = 242\ kJ/mol$
The bond dissociation energy of HCl in the above reaction will be:
1. 232 kJ/mol
2. 331 kJ/mol
3. 431 kJ/mol
4. 530 kJ/mol

69. Calculate the magnitude of heat **(q)** in joules for an isothermal irreversible expansion, where the system expands against an external pressure of **8 bar**, and the volume increases by **10 L**.

1.	8000 J	2.	2000 J
3.	6000 J	4.	7600 J

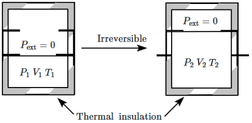
70. Which of the following statements is true for a spontaneous process?

1.	The enthalpy change of the system must be negative.
2.	The entropy change of the system must be positive.
3.	The entropy change of the surroundings must be positive.
4.	The entropy change of the system plus its surroundings must be positive.

71. For the reaction:
$X_2O_4(l) \rightarrow 2XO_2(g)$
with the given values $\Delta U = 2.1\ kcal$ and $\Delta S = 20\ cal\ K^{-1}$ at 300 K, what is the value of ΔG?
1. +2.7 kcal
2. –2.7 kcal
3. +9.3 kcal
4. –9.3 kcal

72. The amount of heat required to raise the temperature of one mole of the substance by 1 K is called its:
1. Molar heat capacity
2. Entropy
3. Thermal capacity
4. Specific heat

73. An ideal gas in a thermally insulated vessel at internal pressure = P_1, volume = V_1, and absolute temperature = T_1 expands irreversibly against zero external pressure, as shown in the diagram. The final internal pressure, volume, and absolute temperature of the gas are P_2, V_2 and T_2, respectively.



Which of the following relation(s) are correct for this expansion? (a) $q = 0$
(b) $T_2 = T_1$
(c) $P_2V_2 = P_1V_1$
(d) $P_2V_2^{\gamma} = P_1V_1^{\gamma}$
1. (a), (b) and (c)
2. (b) and (d)
3. (a) and (d)
4. None of the above

74. The enthalpy of formation of $CO_{(g)}$, $CO_{2(g)}$, $N_2O_{(g)}$, and $N_2O_{4(g)}$ are –110 kJ mol^{-1}, – 393 kJ mol^{-1}, 81 kJ mol^{-1}, and 9.7 kJ mol^{-1} respectively. The value of $(\Delta)_r H$ for the following reaction would be:
$N_2O_{4(g)} + 3CO(g) \rightarrow N_2O_{(g)} + 3CO_{2(g)}$

1.	$-777.7\ kJ\ mol^{-1}$	2.	$+777.7\ kJ\ mol^{-1}$
3.	$+824.9\ kJ\ mol^{-1}$	4.	$-345.4\ kJ\ mol^{-1}$

75. How much work is done by an ideal gas when it expands from 10 L to 50 L under a constant pressure of 2 bar in a single step?
1. W = −8 kJ
2. W = −18 kJ
3. W = −12 kJ
4. W = +8 kJ

76. The value of $\log_{10} K$ for a reaction $A \rightleftharpoons B$ is:
(Given : $\Delta_r H_{298\ K} = -54.07$ kJ mol^{-1},
$\Delta_r S^{\circ}_{298\ K} = 10 JK^{-1}\ mol^{-1}$ and $R = 8.314 JK^{-1}\ mol^{-1}$,
$2.303 \times 8.314 \times 298 = 5705$)
1. 5
2. 10
3. 95
4. 100

77. What is the ΔH (enthalpy change) for the reaction represented by the equation,
$OF_2 + H_2O \rightarrow O_2 + 2HF$?
(Given the bond energies of $O-F, O-H, H-F$ and and $O=O$ as 44, 111, 135, and 119 kcal mol^{-1}, respectively)

1.	–222 kcal	2.	–88 kcal
3.	–111 kcal	4.	–79 kcal

78. The work done when 1 mole of gas expands reversibly and isothermally from a pressure of 5 atm to 1 atm at 300 K is:
(Given log 5 = 0.6989 and R = 8.314 J K^{-1} mol^{-1})
1. zero J
2. 150 J
3. +4014.6 J
4.-4014.6 J

79. The correct thermodynamic conditions for a spontaneous reaction at all temperatures is:
1. ΔH > 0 and ΔS< 0
2. ΔH < 0 and ΔS> 0
3. ΔH < 0 and ΔS< 0
4. ΔH > 0 and ΔS = 0

80. The reaction, among the following, that is anticipated to have the highest change in entropy (ΔS) is:
1. Ca(S) + ½ O_2(g) → CaO(S)
2. C(S) + O_2(g) → CO_2 (g)
3. N_2(g) + O_2(g) → 2NO(g)
4. $CaCO_3$(S) → CaO(S) + CO_2(g)

81. If a reaction is non-spontaneous at the freezing point of water but is spontaneous at the boiling point of water, then:

	ΔH	ΔS
1.	+ve	+ve
2.	-ve	-ve
3.	-ve	+ve
4.	+ve	-ve

82. What is the amount of work done by an ideal gas, if the gas expands isothermally from $10^{-3}\ m^3$ to $10^{-2}\ m^3$ at 300 Kagainst a constant pressure of $10^5\ Nm^{-2}$?

1.	$+270\ kJ$	2.	$-900\ J$
3.	$+900\ kJ$	4.	$-900\ kJ$

83. Considering entropy (S) as a thermodynamic parameter, the criterion for the spontaneity of any process is:

1. $\Delta S_{system} + \Delta S_{surroundings} > 0$
2. $\Delta S_{system} - \Delta S_{surroundings} > 0$
3. $\Delta S_{system} > 0$ only
4. $\Delta S_{s\,\text{urroundings}} > 0$ only

84. Which of the following options correctly describes the free expansion of an ideal gas under adiabatic conditions?
1. $\mathrm{q} \neq 0,\ \ \Delta\mathrm{T} = 0,\ \ \mathrm{W} = 0$
2. $\mathrm{q} = 0,\ \ \Delta\mathrm{T} = 0,\ \ \mathrm{W} = 0$
3. $\mathrm{q} = 0,\ \ \Delta\mathrm{T} < 0,\ \ \mathrm{W} \neq 0$
4. $\mathrm{q} = 0,\ \ \Delta\mathrm{T} \neq 0,\ \ \mathrm{W} = 0$

85. For the reaction of cyanamide $NH_2CN(g)$ with O_2 in a bomb calorimeter, ΔU was found to be -740 kJ/mol at 300 K. The enthalpy change for the reaction $\mathrm{NH_2CN\ (g) + \frac{3}{2}O_2\ (g) \rightarrow N_2\ (g) + CO_2\ (g) + H_2O\ (l)}$ will be:
1. –507.1 kJ /mol
2. –1987.1 kJ/mol
3. –738.75 kJ/mol
4. –741.2 kJ/mol

86. ΔH^o and ΔS^o values for a particular reaction are –60.0 kJ and –0.200 kJ $\mathrm{K^{-1}}$ respectively. Under what conditions is this reaction spontaneous:
1. $T < 300K$
2. $T = 300K$
3. $T > 300K$
4. All of the above

87. What is the value of ΔH for the reaction $2Al + Cr_2O_3 \rightarrow 2Cr + Al_2O_3$, knowing that the enthalpies of formation of Al_2O_3 and Cr_2O_3 are –1596 kJ and –1134 kJ, respectively?

1.	–1365 kJ	2.	2730 kJ
3.	–2730 kJ	4.	–462 kJ

88. 5 moles of an ideal gas at 100 K are allowed to undergo reversible compression till its temperature becomes 200 K. If $\mathrm{C}_V = 28\ \mathrm{J\ K^{-1}} mol^{-1}$, calculate $\Delta\mathrm{U}$ and $\Delta(\mathrm{pV})$for this process. $\left(\mathrm{R} = 8.0\ \mathrm{J\ K^{-1}\ mol^{-1}}\right)$
1. $\Delta\mathrm{U} = 14\ \mathrm{kJ}; \Delta(\mathrm{pV}) = 18\ \mathrm{kJ}$
2. $\Delta\mathrm{U} = 14\ \mathrm{kJ}; \Delta(\mathrm{pV}) = 0.8\ \mathrm{kJ}$
3. $\Delta\mathrm{U} = 14\ \mathrm{kJ}; \Delta(\mathrm{pV}) = 4\ \mathrm{kJ}$
4. $\Delta\mathrm{U} = 14\ \mathrm{kJ}; \Delta(\mathrm{pV}) = 8.0\ \mathrm{kJ}$

89. Consider the following four statements:

I:	Positive entropy change is possible at any temperature.
II:	The combustion of methane is an endothermic process.
III:	q = 0 for a cyclic process.
IV:	q = 0 for an adiabatic process.

Which of the above statements are correct?
1. **III** and **IV**
2. **II**, **III** and **IV**
3. **I**, **III** and **IV**
4. **IV**

90. Select the correct option based on the statements below:

Statement I:	Pressure, molar heat capacity, density, mole fraction, specific heat, temperature, and molarity are intensive properties.
Statement II:	Mass, internal energy, and heat capacity are extensive properties.

1. Both **I** and **II** are true.
2. **I** is true and **II** is false.
3. Both **I** and **II** are false.
4. **I** is false but **II** is true.

Biology

91. Consider the following statements about the stages of prophase I in meiosis:

(i)	Leptotene stage is marked by the gradual visibility of chromosomes under the light microscope.
(ii)	Zygotene stage involves the pairing of homologous chromosomes, forming synaptonemal complexes.
(iii)	Pachytene stage is where the crossing over between non-sister chromatids occurs.

Which of the above statements are correct?
1. Only **(i)** and **(ii)**
2. Only **(ii)** and **(iii)**
3. Only **(i)** and **(iii)**
4. All **(i)**, **(ii)** and **(iii)**

92. Both tendons and ligaments:

I:	are dense irregular connective tissues
II:	connect muscle to bone

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

93. Under normal physiological conditions, every 100 ml of oxygenated blood can deliver around:
1. 4 ml of O_2 to the tissues.
2. 5 ml of O_2 to the tissues.
3. 4 ml of O_2 to the alveoli.
4. 5 ml of O_2 to the alveoli.

94. What is the primary structural precursor to the new cell wall formed during plant cytokinesis?
1. Cell plate
2. Middle lamella
3. Spindle fibres
4. Metaphase plate

95. Dicotyledonae represents
a. Class
b. Order
c. Division
d. Genus

96. Match Column I and Column II with respect to transport of CO_2 and O_2 in different forms with their percentages:

	Column I		**Column II**
(a)	CO_2 as carbonates by plasma	(i)	23%
(b)	CO_2 as Carbaminohaemoglobin by RBCs	(ii)	3%
(c)	CO_2 in dissolved form by plasma	(iii)	70%
(d)	O_2 in dissolved form by plasma	(iv)	7%

1. (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)
2. (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
3. (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
4. (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)

97. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A):	With the help of several ommatidia, a cockroach can perceive several images of an object, i.e., mosaic vision
Reason (R):	Mosaic vision gives less sensitivity but more resolution

In the light of the above statements, choose the most appropriate answer from the options given below:

1.	**(A)** is correct but **(R)** is not correct
2.	**(A)** is not correct but **(R)** is correct
3.	Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**
4.	Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

98. Match List-I with List-II:

	List-I		**List-II**
A.	Residual volume	I.	Maximum volume of air that can be breathed in after forced expiration
B.	Vital capacity	II.	Volume of air inspired or expired during normal respiration
C.	Expiratory capacity	III.	Volume of air remaining in lungs after forcible expiration
D.	Tidal Volume	IV.	Total volume of air expired after normal inspiration

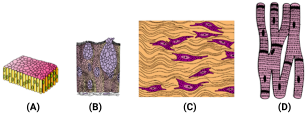
Choose the correct answer from the options given below:
1. A-IV,B-III,C-II,D-I
2. A-II,B-IV,C-I,D-III
3. A-III,B-I,C-IV,D-II
4. A-I,B-II,C-III,D-IV

99. Cells in our body, that are not dividing, are said to be at which phase of the cell cycle?

1.	G_1	2.	G_2
3.	G0	4.	S phase

100. By increasing the area of observation…..and ……… of organisms will increase.
1. Size and variety
2. Range and variety
3. Efficiency and range
4. Growth and variety

101. The four sketches (A, B, C, and D) given below, represent four different types of animal tissues. Which one of these is correctly identified in the options given, along with its correct location and function?



		Tissue	Location	Function
1.	(C)	Collagen fibres	Cartilage	Attach skeletal muscles to bones
2.	(D)	Smooth muscle tissue	Heart	Heart contraction
3.	(A)	Columnar epithelium	Nephron	Secretion and absorption
4.	(B)	Glandular epithelium	Intestine	Secretion

102. Which of the following statements best explains why CO_2 transport in blood occurs primarily in the form of bicarbonate ions?

1.	CO_2 has higher solubility in plasma, allowing efficient dissolution and transport.
2.	Carbonic anhydrase in red blood cells rapidly converts CO_2 to bicarbonate ions.
3.	Hemoglobin has a higher affinity for bicarbonate ion than for CO_2.
4.	Plasma proteins facilitate CO_2 transport as bicarbonate ions.

103. Polymoniales includes-
1. Convolvulaceae
2. Solanaceae
3. Both 1 and 2
4. Liliaceae only

104. Which of the following statement is incorrect?

1.	Taxa can indicate categories at very different levels.
2.	systematics takes into account evolutionary relationship between organisms.
3.	Modern taxonomy do not take into account ecological information about organisms.
4.	Each rank or taxon represents unit of classification.

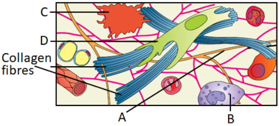
105. In inspiration during pulmonary ventilation the correct order of the given events is:
a. air flows into the lungs
b. alveolar volume increases
c. thoracic volume increases
d. pleural pressure decreases
e. alveolar pressure decreases

1.	a, b, c, d, e	2.	b, e, a, c, d
3.	c, d, b, e, a	4.	d, e, b, a, c

106. In an animal cell, cytokinesis is achieved by:

1.	the appearance of a furrow in the plasma membrane which gradually deepens and ultimately joins in the centre dividing the cell cytoplasm into two.
2.	the appearance of the cell-plate and the formation starts in the centre of the cell and grows outward to meet the existing lateral walls.
3.	the appearance of a furrow in the plasma membrane which starts in the centre of the cell and grows outward to meet the existing lateral walls.
4.	the appearance of the cell-plate and the formation starts in the periphery of the cell and grows inward to meet the existing central walls.

107. In the given diagram of areolar tissue, identify a phagocytic cell:



1.	A	2.	B
3.	C	4.	D

108. Which of the following sense organs in frogs are well-organised structures rather than cellular aggregations around nerve endings?

I:	Sensory papillae for touch
II:	Taste buds
III:	Nasal epithelium
IV:	Eyes
V:	Tympanum with internal ears

1.	Only **I**, **II** and **III**	2.	Only **IV** and **V**
3.	Only **II**, **IV** and **V**	4.	Only **IV**

109. Oxyhaemoglobin formation and dissociation depend on several physiological factors. Under which condition does oxyhaemoglobin dissociate more readily?

1.	High pO_2, low pCO_2, low H^+ concentration, and low temperature
2.	High pO_2, high pCO_2, high H^+ concentration, and high temperature
3.	Low pO_2, high pCO_2, high H^+ concentration, and high temperature
4.	Low pO_2, low pCO_2, high H^+ concentration, and low temperature

110. What is the best description of an endocrine gland?

1.	They secrete mucus, saliva, ear wax and digestive enzymes
2.	Their products are released through ducts or tubes
3.	Unlike exocrine glands, they are not modified epithelium
4.	Their secretions are secreted directly into the fluid bathing the gland

111. What is a characteristic feature of the G1 phase of the cell cycle?

1.	DNA replication occurs.
2.	Chromosomes are visible under a light microscope.
3.	The cell undergoes division.
4.	The cell grows in size and synthesizes mRNA and proteins.

112. The total amount of air that can be forcibly exhaled after maximum inhalation measures:
1. Tidal Volume
2. Inspiratory Reserve Volume
3. Residual Volume
4. Vital Capacity

113. In which of the following connective tissues, the cells secrete fibres of collagen or elastin?
A. Cartilage
B. Bone
C. Adipose tissue
D. Blood
E. Areolar tissue
Choose the most appropriate answer from the options given below:
1. **B**,**C**,**D** and **E** only
2. **A**,**B**,**C** and **E** only
3. **B**,**C** and **D** only
4. **A**,**C** and **D** only

114. Consider the given two statements:

Assertion (A):	Metaphase is the stage at which morphology of chromosomes is most easily studied.
Reason (R) :	At metaphase, condensation of chromosomes is completed.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	**(A)** is False but **(R)** is True.

115. Telophase is the final stage of karyokinesis where two daughter nuclei form. Which of the following cellular events specifically marks this stage?

1.	Chromosomes condense into tightly packed structures for mitotic completion.
2.	The nuclear envelope reforms around decondensing chromatin, and nucleoli reappear.
3.	Chromatids separate and move towards opposite spindle poles.
4.	DNA replication takes place in preparation for the next division.

116. During Anaphase I of Meiosis I:

1.	sister chromatids move toward opposite poles
2.	homologous chromosomes move toward opposite poles
3.	homologues move toward the same pole
4.	homologous chromosomes move randomly toward either pole

117. The distal convoluted tubule of the human nephron is lined with:
1. Simple squamous epithelium
2. Simple cuboidal epithelium
3. Simple columnar epithelium
4. Stratified squamous epithelium

118. indica is-

1.	Specific epithet of mango
2.	Specific epithet of Indus valley species
3.	Specific epithet of wheat (indian species)
4.	Specific epithet of Rice (indian species)

119. Which of the following factors does not favor the release of oxygen from hemoglobin at the tissues?
1. High partial pressure of CO_2 (pCO_2).
2. Low pH (acidic conditions).
3. High temperature.
4. High partial pressure of O_2 (pO_2).

120.
Given below are two statements:

Assertion (A):	Cats and dogs have some similarities
Reason (R):	Cats and dogs belong to the same family canidae

1.	Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is true but **(R)** is false.
4.	Both **(A)** and **(R)** are false.

121. Given below are two statements:

Assertion (A):	'Animals' , 'mammals' and 'dogs' represent taxa at different levels
Reason (R):	Different taxa can never occupy same hierarchy

1.	Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is true but **(R)** is false.
4.	Both **(A)** and **(R)** are false.

122. Which statement is true regarding the events of the G1, S, and G2 phases of the cell cycle?

1.	DNA replication occurs in the G1 phase.
2.	The cell grows and synthesizes proteins necessary for mitosis in the S phase.
3.	The cell checks for DNA errors and begins to form the necessary components for cell division in the G0 phase.
4.	DNA is synthesized in the S phase.

123. Match each item in Column I with ONLY one in Column II and select the correct match from the codes given:

	Column I [Structure in Cockroach]		**Column II [Location]**
A	Forewings	P	Prothorax
B	Hind wings	Q	Mesothorax
C	Anal cerci	R	Metathorax
		S	10^{th} segment but absent in females
		T	10^{th} segment and present in both sexes

Codes:

	A	**B**	**C**
1.	Q	R	S
2.	Q	R	T
3.	P	Q	S
4.	R	R	T

124. Kinetochores:

I:	are small disc-shaped structures at the surface of the centromeres.
II:	serve as the sites of attachment of spindle fibres to the chromosomes.
III:	are same as centromeres.

1. **I** is correct; **II** & **III** are incorrect
2. **I** & **II** are correct; **III** is incorrect
3. **I**, **II** & **III** are correct
4. **I**, **II** & **III** are not correct.

125. Match the phases of Prophase I in **Column-I** with corresponding events in **Column-II** and select the correct match from the codes given:

	Column-I		Column-II
A.	Zygotene	P.	Recombination
B.	Pachytene	Q.	Terminalisation of chismata
C.	Diplotene	R.	Synapsis
D.	Diakinesis	S.	Dissolution of synaptonemal complex

Codes:

	A	B	C	D
1.	R	P	S	Q
2.	R	P	Q	S
3.	P	R	S	Q
4.	P	R	Q	S

126. In animal cells, during the S phase:

I:	DNA replication begins in the cytoplasm
II:	The centriole duplicates in the nucleus

1.	Only **I** is correct
2.	Only **II** is correct
3.	Both **I** and **II** are correct
4.	Both **I** and **II** are incorrect

127. Scientific name does not ensure

1.	Each organism has only one name
2.	Description of any organism lead to the same name of organism in any part of the world
3.	No two organisms have the same name
4.	Status of threat of extinction of that organism holding a specific scientific name.

128. Match the stages of prophase I in **Column-I** with their corresponding events in **Column-II** and select the best match
from the codes given:

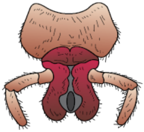
	Column I (Stages of Prophase I)		Column II (Events)
A.	Leptotene	P.	Terminalization of chiasmata and breakdown of the nuclear envelope.
B.	Zygotene	Q.	Dissolution of synaptonemal complex and visible chiasmata form.
C.	Pachytene	R.	Chromosomes become gradually visible and condensation begins.
D.	Diplotene	S.	Crossing over occurs at recombination nodules.
E.	Diakinesis	T.	Homologous chromosomes pair and form synaptonemal complexes.

Options	A	B	C	D	E
1	R	T	S	Q	P
2	T	R	Q	S	P
3	R	S	T	Q	P
4	P	Q	R	S	T

129. ICBN stands for

1.	International Code for Binomial Nomenclature
2.	International Code for Biological Nomenclature
3.	International Code for Botanical Nomenclature
4.	International Code for Biogeographical Nomenclature

130. The mouth part of Cockroach shown in the given figure:



1. is paired
2. acts as the tongue
3. forms the upper lip
4. forms the lower lip

131. Lungs do not collapse between breaths and some air always remains in the lungs which can never be expelled because?

1.	there is a negative pressure in the lungs
2.	there is a negative intrapleural pressure pulling at the lung walls
3.	there is a positive intrapleural pressure
4.	pressure in the lungs is higher than the atmospheric pressure

132. Many parts of frog brains correspond with those of humans. Which structure of human brain has no corresponding structure in frogs?

1.	Olfactory lobe	2.	Optic lobe
3.	Pons	4.	Cerebellum

133. Which one of the following animals is correctly matched with its particular named taxonomic category?

1.	Cuttlefish	–	Mollusca, a class
2.	Humans	–	Primata, the family
3.	Housefly	–	*Musca*, an order
4.	Tiger	–	*tigris*, the species

134. Select the option that does not represent the taxa at different levels

1.	Chordate, Mammal, Dog
2.	Animal, Insect, Butterfly
3.	Plant, Monocot, Wheat
4.	Mammal, Insect, Dicot

135. Consider the given two statements:

Statement I:	Ligaments connect bones to bones and contain a high proportion of elastic fibres.
Statement II:	Tendons connect muscles to bones and consist mainly of collagen fibers, providing great strength but limited elasticity.

1.	**Statement I** is correct; **Statement II** is correct
2.	**Statement I** is correct; **Statement II** is incorrect
3.	**Statement I** is incorrect; **Statement II** is correct
4.	**Statement I** is incorrect; **Statement II** is incorrect

136. Given below are two statements:

I:	Goblet cells are unicellular glands
II:	Earwax is the secretion of exocrine gland

In the light of the above statements, choose the correct answer from the options given below:
1. **Statement I** is True but **Statement II** is False
2. **Statement I** is False but **Statement II** is True
3. Both **Statement I** and **Statement II** are True
4. Both **Statement I** and **Statement II** are False

137. As each chromosome moves away from the equatorial plate during the anaphase of mitosis:

1.	the arms of each chromosome remain directed towards the pole and hence at the leading edge, with the centromere of the chromosome trailing behind.
2.	the arms of each chromosome and the centromere remain directed towards the pole and hence at the leading edge.
3.	the centromere of each chromosome remains directed towards the pole and hence at the leading edge, with the arms of the chromosome trailing behind.
4.	the centromere and the arms of each chromosome do not move and remain at the metaphase plate.

138. Regarding the respiratory system of cockroach:

I:	It consists of a network of trachea that open through 12 pairs of spiracles present on ventral side of the body.
II:	The opening of spiracles is regulated by sphincters.
III:	Exchange of gases takes place at the tracheoles by diffusion.

1. Only **I** and **II** are correct
2. Only **I** and **III** are correct
3. Only **II** and **III** are correct
4. **I, II** and **III** are correct

139. Inflammation of bronchi and bronchioles leading to difficulty in breathing is characteristically seen in:
1. Bronchial asthma
2. Emphysema
3. Tuberculosis
4. Pneumonia

140. Consider the following statements:

I.	Higher the taxa, more are the characteristics that members whithin the taxon share.
II.	Lower the category, greater is the difficulty of determining the relationship to other taxa at the same level.

Of the two statements:

1.	**I** is true but **II** is false
2.	**I** is false but **II** is true
3.	Both **I** and **II** are true
4.	Both **I** and **II** are false

141. Trachea is a straight tube extending up to the mid-thoracic cavity, which divides into a right and left primary bronchi at the level corresponding to:
1. Seventh cervical vertebra
2. Second thoracic vertebra
3. Fifth thoracic vertebra
4. Seventh thoracic vertebra

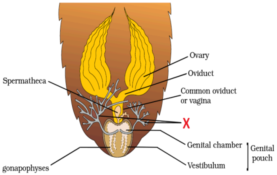
142. Under normal physiological conditions every 100 ml of deoxygenated blood delivers:
1. approximately 4 ml of CO_2 to the alveoli.
2. approximately 5 ml of CO_2 to the alveoli.
3. approximately 4 ml of CO_2 to the tissues.
4. approximately 5 ml of CO_2 to the tissues.

143. Consider the following statements about the digestive system of frogs:

A:	The alimentary canal is short because frogs are herbivores.
B:	The liver secretes bile, which is stored in the gall bladder.
C:	The pancreas produces digestive enzymes that are released into the duodenum.

Which of the above statements is/are incorrect?
1. **A** only
2. **B** and **C** only
3. **A** and **B** only
4. **B** only

144. What is 'X' in the given diagram?



1.	Collateral Gland	2.	Phallic Gland
3.	Mushroom Gland	4.	Corpora Allata

145. Latin origin is indicated by

1.	By separately underlining when handwritten
2.	By printing in italics
3.	Both a and b
4.	By capitalizing first alphabet of generic name.

146. Consider the given two statements:

Assertion (A):	O_2 has a greater diffusing capacity than CO_2 across the respiratory diffusion membrane.
Reason (R):	O_2 is more soluble in water than CO_2.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**
3.	**(A)** is True but **(R)** is False
4.	**Both (A)** and **(R)** are False

147. Given below are two statements:

Statement I:	Ligaments are dense irregular tissue.
Statement II:	Cartilage is dense regular tissue.

In the light of the above statements, choose the correct answer from the options given below:

1.	**Statement I** is false but **Statement II** is true.
2.	Both **Statement I** and **Statement II** are true.
3.	Both **Statement I** and **Statement II** are false.
4.	**Statement I** is true but **Statement II** is false.

148. Consider the given two statements:

Assertion (A):	During inspiration, the intra-pulmonary pressure becomes less than the atmospheric pressure.
Reason (R):	Contraction of the diaphragm and external intercostal muscles increases the thoracic volume.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	**(A)** is False but **(R)** is True.

149. Most digestion in frog occurs in:
1. Oesophagus
2. Stomach
3. Small intestine
4. Rectum

150. Consider the given two statements:

Statement I:	In the alveoli, conditions such as high pO_2, low pCO_2, lesser H^+ concentration, and lower temperature favor the formation of oxyhaemoglobin.
Statement II:	In the tissues, where there is low pO_2, high pCO_2, high H^+ concentration, and higher temperature, the dissociation of oxygen from oxyhaemoglobin is favored.

1.	Both **Statement I** and **Statement II** are correct, and **Statement II** correctly explains **Statement I**.
2.	Both **Statement I** and **Statement II** are correct, but **Statement II** does not explain **Statement I**
3.	**Statement I** is correct, but **Statement II** is incorrect.
4.	**Statement I** is incorrect, but **Statement II** is correct.

151. The trachea:

I:	divides into two principal bronchi at the level of 7th thoracic vertebra.
II:	is invested by incomplete cartilaginous rings in its wall.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

152. Classes' together form……...in plants.
a. Phylum
b. Division
c. Kingdom
d. Order

153. What is prerequisite for nomenclature?

1.	Description of organism
2.	Knowledge of which organism is considered for a particular name
3.	Identification
4.	All are correct

154. Consider the two statements:

Statement I:	Peripheral chemoreceptors can recognise changes in CO_2 and H^+ concentration
Statement II:	The role of oxygen in the regulation of respiratory rhythm is quite insignificant

1.	**Statement I** is correct and **Statement II** is incorrect
2.	**Statement I** is incorrect and **Statement II** is correct
3.	Both **Statement I** and **Statement II** are correct
4.	Both **Statement I** and **Statement II** are incorrect

155. Contraction of diaphragm:

1.	increases the diameter of thorax in dorso-ventral axis and brings about inhalation
2.	decreases the diameter of thorax in dorso-ventral axis and brings about exhalation
3.	increases the diameter of thorax in antero-posterior axis and brings about inhalation
4.	decreases the diameter of thorax in antero-posterior axis and brings about exhalation

156. The institute which encourages publication of local flora in India is
1. NBRI
2. FRI
3. BSI
4.IARI

157. The respiratory rhythm centre is located in:
1. Cerebrum.
2. Cerebellum.
3. Pons.
4. Medulla.

158. The terms for ‘summer sleep’ and ‘winter sleep’ seen in frogs respectively are:

1.	Aestivation and Hibernation
2.	Hibernation and Aestivation
3.	Diapause and Cryptobiosis
4.	Cryptobiosis and Diapause

159. Which of the following statements is not correct with respect to the epithelial tissue?

1.	It covers the external surface of the body and the internal surface of many organs.
2.	The neighbouring cells are held together by cell junctions and there is very little extracellular material.
3.	The epithelial cells rest on a cellular basement membrane that separates it from underlying connective tissue.
4.	There is no blood vessel supplying the nutrients to the epithelial cells.

160. Consider the two given statements:

Statement I:	At the time of cytoplasmic division, organelles like mitochondria and plastids get distributed between the two daughter cells.
Statement II:	In some organisms karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises leading to the formation of syncytium (e.g., liquid endosperm in coconut).

1. **Statement I** is incorrect; **Statement II** is incorrect
2. **Statement I** is correct; **Statement II** is incorrect
3. **Statement I** is incorrect; **Statement II** is correct
4. **Statement I** is correct; **Statement II** is correct

161. In meiotic cell division, the simultaneous splitting of the centromere of each chromosome allowing sister chromatids to move toward opposite poles of the cell by shortening of microtubules attached to kinetochores, occurs during:
1. Metaphase I
2. Metaphase II
3. Anaphase I
4. Anaphase II

162. The events of a cell cycle:

I:	are themselves under genetic control
II:	have to take place in a coordinated way

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

163. During prophase I of meiosis I, recombination between homologous chromosomes is completed:
1. in the Zygotene
2. in the beginning of the Pachytene
3. by the end of Pachytene
4. in the beginning of the Diplotene

164. In male cockroaches, the genital pouch is bounded by:

1.	9^{th} and 10^{th} terga dorsally and 10^{th} sternum ventrally
2.	9^{th} tergum dorsally and 9^{th} and 10^{th} sterna ventrally
3.	9^{th} and 10^{th} terga dorsally and 9^{th} sternum ventrally
4.	9^{th} and 10^{th} terga dorsally and 9^{th} and 10^{th} sterna ventrally

165. Nomenclature is
a. assigning a single name to an organism valid all over the world.
b. Defining properties of an organism
c. Localizing the area of abundance for that organism
d. Assigning three sets of name applicable at different levels of classification.

166. Match the following cell structure with its characteristic feature:

	Column I		Column II
(a)	Tight junctions	(i)	Cement neighboring cells together to form a sheet
(b)	Adhering junctions	(ii)	Transmit information through chemicals to another cell
(c)	Gap junctions	(iii)	Establish a barrier to prevent leakage of fluid across epithelial cells
(d)	Synaptic junctions	(iv)	Cytoplasmic channels to facilitate communication between adjacent cells

Select the correct option from the following:

Options:	(a)	(b)	(c)	(d)
1.	(ii)	(iv)	(i)	(iii)
2.	(iv)	(ii)	(i)	(iii)
3.	(iii)	(i)	(iv)	(ii)
4.	(iv)	(iii)	(i)	(ii)

167. In plant cells:

I:	during the S phase, DNA replication begins in the nucleus.
II:	during the S phase, the centriole duplicates in the cytoplasm.
III:	during the G_2 phase, proteins are synthesised in preparation for mitosis
IV:	cell growth stops during S phase and G_2 phase.

1. **I** is correct; **II**, **III** & **IV** are incorrect
2. **I** & **IV** are correct; **II** & **III** are incorrect
3. **I** & **III** are correct; **II** and **IV** are incorrect
4. All the statements: **I**, **II**, **III** & **IV** are correct.

168. Identify the incorrect statement regarding carbonic anhydrase:

1.	RBCs contain a very high concentration of the enzyme.
2.	It is not present at all in the plasma.
3.	It assists rapid inter-conversion of carbon dioxide and water into carbonic acid, protons and bicarbonate ions.
4.	The active site of the enzyme contains a zinc ion.

169. The lungs are covered by the pleural membranes which mainly help in:
1. maintaining pressure
2. increasing thoracic pressure
3. removing foreign particles
4. reducing friction

170. Sapindales is which taxon?

1.	Order	2.	Class
3.	Genus	4.	Species

171. Which of the following is/are referred to as unit of classification?

(a)	Category
(b)	Organism
(c)	Taxon
(d)	Rank

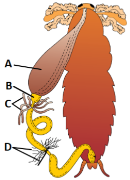
1.	**(a)** only
2.	All **(a), (b), (c)** and **(d)**
3.	**(a), (c)** and **(d)**
4.	**(a)** and **(c)** only

172. The enzyme recombinase is required at which stage of meiosis:

1.	Pachytene	2.	Zygotene
3.	Diplotene	4.	Diakinesis

173. The suffix that denotes the taxonomic category 'Family' in plants would be
(1) -ae
(2) -aceae
(3) -ales
(4) -nae

174. In the following diagram of alimentary canal of cockroach, identify the labels A, B, C and D respectively.



1.	A-Gizzard, B-Crop, C-Malpighian tubules, D-Hepatic caeca
2.	A-Oesophagus, B-Crop, C-salivary glands, D-Hepatic caeca
3.	A-Crop, B-Gizzard, C-Hepatic caeca, D-Malpighian tubules
4.	A-Gizzard, B-Hepatic caeca, C-Malpighian tubules, D- Cilia

175. Which of the following is a correct match between the biological name and the order of the organism?
1. *Homo sapiens* – Hominidae
2. *Mangifera indica* - Anacardiaceae
3. *Triticum aestivum* - Poaceae
4. *Mangifera indica* - Sapindales

176. Chemosensitive area situated adjacent to the rhythm center is highly sensitive to
A: Carbon dioxide
B: Oxygen
C: Hydrogen ions
1. Only **A** and **B**
2. Only **A and C**
3. Only **B and C**
4. **A**, **B** and **C**

177. When compared with carbon dioxide, the partial pressure of oxygen in the air and its solubility in water is respectively:

1.	lower, lower	2.	lower, higher
3.	greater, lower	4.	greater, higher

178. Consider the following statements regarding the G0 phase:

(i)	Cells in the G_0 phase are metabolically inactive.
(ii)	Cells in the G_0 phase exit the G_1 phase and do not divide further.
(iii)	G_0 phase is also known as the quiescent stage of the cell cycle.

Which of the above statement(s) is/are correct?
1. Only **(ii)** and **(iii)**
2. Only **(i)** and **(iii)**
3 Only **(i)** and **(ii)**
4. Only **(iii)**

179.
Match the following entities of column I with their respective orders of column II and choose the correct combination from the option.

	Column I		Column II
A.	Wheat	1.	Primate
B.	Mango	2.	Diptera
C.	Housefly	3.	Sapindales
D.	Man	4.	Poales

1. A-4 B-3 C-2 D-1
2. A-1 B-2 C-4 D-3
3. A-3 B-4 C-2 D-1
4. A-2 B-4 C-1 D-3

180. In the 24 hour average duration of cell cycle of a human cell, cell division properly lasts for:

1.	only about 1 hour
2.	approximately 90 minutes
3.	between 4 and 5 hours
4.	95% time of the cell cycle